

SOLUTIONS TO EXERCISES OF FIBER BUNDLEFROM LIANG'S TEXTBOOK: DIFFERENTIAL GEOMETRY AND GENERAL RELATIVITY

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ABSTRACT. There is a well-known textbook in Chinese theoretic physics community called *Differential Geometry and General Relativity* by Liang and Zhou and in Volumn III of the series, there is a chapter on Fiber Bundle. This chapter is a good introduction to the concept of Fiber Bundle for us as beginners and is a good supplement to the book. Here I provide the solutions to the exercises in the chapter.

1. DEFINITIONS AND THEOREMS

Definition 1.1. *Principal fiber bundle* consists of a *bundle manifold* P , a *base manifold* M and a *structure Lie group* G . They satisfy the following conditions:

- (1) G acts freely on P from the right via the action $R : P \times G \rightarrow P$;
- (2) There exists a smooth projection map $\pi : P \rightarrow M$ such that for $\forall p \in P$, the fiber over $\pi(p)$ is given by $\pi^{-1}[\pi(p)] = \{pg \mid g \in G\}$;
- (3) $\forall x \in M$, there exists an open neighborhood U of x in M and a diffeomorphism $T_U : \pi^{-1}[U] \rightarrow U \times G$, $T_U(p) = (\pi(p), S_U(p))$, $\forall p \in \pi^{-1}[U]$, where $S_U : \pi^{-1}[U] \rightarrow G$ satisfies $S_U(pg) = S_U(p)g$, $\forall g \in G$.

Definition 1.2. Let $R_p : G \rightarrow P$, $R_p(g) := R_g(p)$, then we can define a fundamental vector field $A_p^* : T_p P \rightarrow \mathfrak{g}$ by $A_p^*(X) := (R_p)_{*p}(X)$, $\forall X \in T_p P$.

The definition of a *connection* on a principal fiber bundle has three equivalent forms:

Definition 1.3. A connection on a principal fiber bundle $P(M, G)$ assigns to each point $p \in P$ a *horizontal subspace* $H_p \subseteq T_p P$ such that:

- (1) $T_p P = V_p \oplus H_p$, $\forall p \in P$, where $V_p := \{X \in T_p P \mid \pi_*(X) = 0\}$ is *vertical subspace*;
- (2) $R_{g*}[H_p] = H_{pg}$, $\forall p \in P$, $\forall g \in G$;
- (3) H_p is C^∞ with respect to p .

Definition 1.4. A connection on a principal fiber bundle $P(M, G)$ is a C^∞ $\mathfrak{g}$ -valued 1-form on P , denoted by $\tilde{\omega}$, such that:

- (1) $\tilde{\omega}_p(A_p^*) = A$, $\forall A \in \mathfrak{g}$, $\forall p \in P$;

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2. PROBLEM 1

Problem 2.1. Prove Eq.(I-1-4), namely $S_U : \pi^{-1}[x] \rightarrow G$ and $R_{\check{p}_U} : G \rightarrow \pi^{-1}[x]$ are a pair of inverse mappings.

Proof.

□

3. PROBLEM 2

Problem 3.1. Prove Eq.(I-1-5), namely $R_g \circ R_p = R_{pf}$ for $\forall p \in P, \forall g \in G$.

Proof.

□

REFERENCES

1. Canbin Liang and Bin Zhou, *Introduction to differential geometry and general relativity (volume iii)*, 2 ed., Science Press, Beijing, 2009.